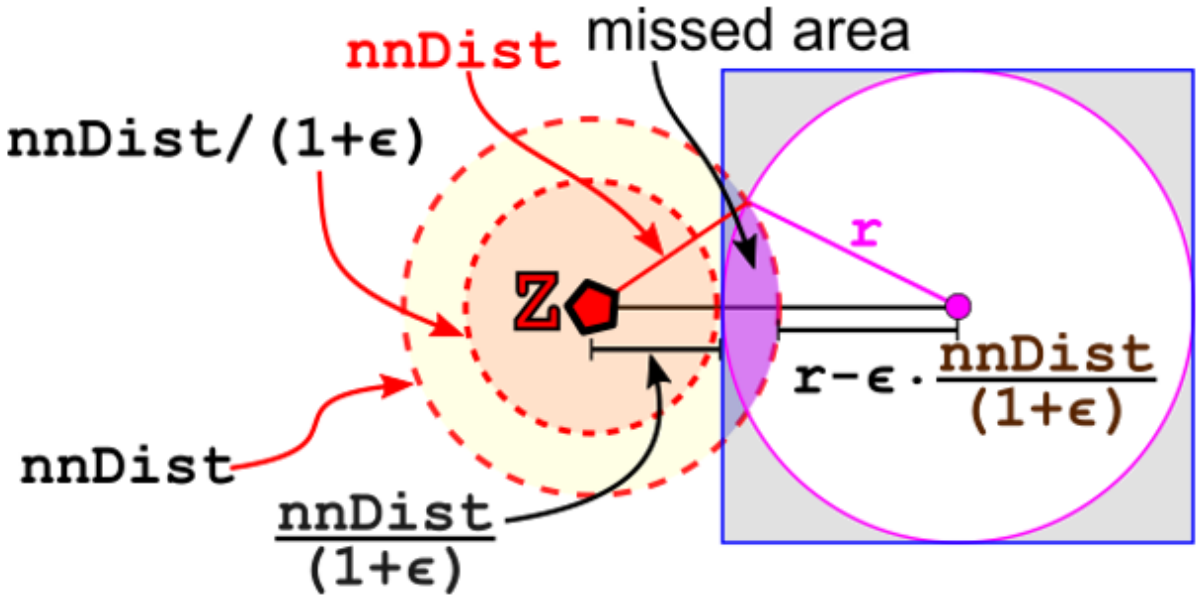


Figure 10.23 The probability of missing `j` points by using an approximation error is proportional to the intersection of the pruned search area (the ring with radius `(ϵ /(1+ϵ))*nnDist`) and the bounding envelope of the node intersecting this region, but not the internal sphere with reduced radius. The maximum probability corresponds to a region tangent to the internal sphere: the figure shows how the intersection is smaller for spherical bounding envelops, with respect to rectangular ones.

If we set `ϵ==0.0`, then we get back the exact search algorithm as a special case, because at line #6 `nnDist/(1+epsilon)` becomes just `nnDist`.

When `ϵ>0.0`, instead, a traversal might look like figure 10.24, where we use an example slightly different from the one in figures 10.20-21, to show how approximate NN search could miss the nearest neighbor.